Vladimir Vladimirovich Baryshev

Neural decoding | Neurotechnology | Biomedical AI | Computer vision

Email | LinkedIn | GitHub | ORCID

PROFILE

Neurotechnology MSc student and Research Collaborator focused on decoding information from neural signals with machine learning. Experience spans intracranial ECoG speech decoding, EEG classification, computer vision, and biomedical image analysis, with hands-on work across research design, data processing, neural-network training, and scientific writing.

RESEARCH AND PROFESSIONAL EXPERIENCE

Research Collaborator - HSE University, Centre for Bioelectric Interfaces

Part-time, remote | Moscow, Russia | Jun 2026 - Present

Work on intracranial speech decoding from human electrocorticography (ECoG). Build and test neural-decoding approaches; preliminary results improve on the group's previous ECoG speech-decoding benchmark.

Independent ML Developer / Researcher

Self-employed | Apr 2026 - Present

Develop research projects in medical image segmentation and EEG classification, including a pelvic CT segmentation support system and an EEGNet classifier on JapanDataset.

Research Assistant / Intern - Skolkovo Institute of Science and Technology, Neurocenter

Research Assistant, part-time: Nov 2025 - Apr 2026 | Intern: Sep 2025 - Nov 2025

Performed neural-data analysis and built Python and computer-vision workflows for neuroscience research.

EDUCATION

MSc, Neurotechnology with the Basics of Biomaterials Science

Sirius University, Sochi | In progress

Master's thesis: spatiotemporal neuromodulation and biomimetic neurointerfaces. Developing an adaptive closed-loop neuromodulation approach for personalized neurorehabilitation.

BSc, General Biology - Molecular Biology and Genetics

North-Caucasus Federal University, Stavropol | GPA 4.35

Developed software for morphological processing of histological slice images and object segmentation; work supported by Russian Science Foundation project 23-71-10013.

PUBLICATION

Lyakhov P. A., Lyakhova U. A., Baboshina V. A., Baryshev V. V., Nagornov N. N. Detection of attention state in children with autism spectrum disorder based on neural network classification of electroencephalograms. Vestnik of Saint Petersburg University. Applied Mathematics. Computer Science. Control Processes, 2025, 21(1), 92-111. Q3 Scopus. DOI

SELECTED PROJECTS

Attention-state detection in children with autism spectrum disorder

EEG classification | Published

Led the full research cycle from problem definition and literature review to EEG preprocessing, neural-network training, collaboration with medical institutions, and paper writing.

Pelvic CT segmentation support system

3D medical imaging | In progress

Build an ML-based clinical decision-support prototype for segmentation of pelvic bones and lower limbs. GitHub

EEGNet classifier on JapanDataset

EEG deep learning | In progress

Develop and evaluate an EEG classifier based on the EEGNet architecture. GitHub

Histological slice processing software

Python desktop application | Completed

Built an end-to-end image-processing application for color correction, channel normalization, threshold binarization, and morphological operations using OpenCV, NumPy, Matplotlib, and PyQt.

TECHNICAL SKILLS

- Programming: Python; basic Git and Linux
- ML and scientific computing: PyTorch, NumPy, MNE, Matplotlib
- Neural and biomedical data: ECoG, EEG, MRI, CT, neural decoding, signal preprocessing
- Computer vision: OpenCV, 3D U-Net, nnU-Net, YOLO, SAM, DeepLabCut
- Application development: PyQt

SELECTED TRAINING

- Neuromatch Academy - Computational Neuroscience, 2025
- New Generation Neurointerfaces, HSE University, 2025
- HSE Brain Research School, 2024
- Computational Neuroscience, University of Washington (Coursera)
- Neuroscience and Neuroimaging Specialization, Johns Hopkins University (Coursera)

SCIENCE COMMUNICATION AND LANGUAGES

Staff science writer at Science Mail with more than 800 popular-science articles across neuroscience, genetics, physics, artificial intelligence, and biomedicine.

- Russian: native
- English: B2 (intermediate-upper intermediate)